

Bet selection: three families of rule, and where the edge is not

Paper 6 in a series on UK All-Weather racing outcomes

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2026-09-10

A betting rule has three parts: which horse to back, which races to bet on the strength of the model's output, and which races to bet on the strength of the race itself. Two earlier rounds searched the first two and found nothing that beat the rule paper 1 took from Owen. This paper adds the third: filters on field size, class, course, distance and going, all known the night before the race. Every candidate in all three families then had to rank in the top decile of its family on both halves of the search window before it could reach the test split. That guard struck Owen's rule on three of the four markets. It did not save the search. One rule topped every search-set table, backing the horse whose model probability most exceeds its market probability in fields of eleven to twelve runners, at +37.2% on the win market with the two halves of the window two points apart. On the test split it returned -38.1%. Search-set rank predicts test rank negatively on all four markets. Two thirds to four fifths of the variation in return between rules is explained by the mean price the rule backs, so a leaderboard mostly ranks price bands, and the residual does not hold across halves of one window. The two best rules on test were the two carried forward without being selected: the market favourite in every race, with no model in it, and Owen's rule. Three rules in the shortlist are profitable once the bookmaker's margin is removed; none is profitable with it in place. Paper 4 showed the model ties the settled starting price and beats the overnight forecast price, so its value sits between them, and every backtest in this series settles at the closing end of that interval. The final section sets out what that implies for live use.

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1 What was searched

A betting rule has three parts: which horse to back, which races to bet on the model’s output, and which races to bet on the race itself. Rounds 1 and 2 searched the first two and found nothing that beat the rule paper 1 took from Owen (2019). That was a claim about two families. This paper adds the third.

Table 1: The five pickers. P5 is Owen’s; with no filter every runner passes its cuts, so P5 is then identical to P1.

Picker	Definition	Reads
P1	Highest model win probability	the field
P2	Market favourite: shortest starting price	the field
P3	Highest model probability $\times$ starting price (expected return)	the field
P4	Largest model probability minus market probability	the field
P5	Highest model probability among runners passing the filter’s cuts	the filter’s cuts

Threshold filters cut on the model probability $\hat{P}$ and the ratio $\hat{P}/P^{\text{mkt}}$. Owen’s rule is $\hat{P} > 0.15$, ratio > 1.3 , picker P5. The grid is $\hat{P}$ from 0.00 to 0.35 by 0.01 crossed with the ratio from 0.80 to 2.50 by 0.05, plus a no-filter row: 1,261 filters.

Race-characteristic filters use conditions known the night before. Every contiguous field-size band over 4 to 16 runners; each class 2 to 5 singly and in contiguous pairs; each course singly and in pairs; the search set’s distance terciles (1540 and 1980 yards) and the cuts at each boundary; the going ordinal at each interior cut. A filter is kept only if it admits 300 search-set races and 150 in each half of the window.

Table 2: The race-characteristic family, before scoring.

Family	Enumerated	Kept	Struck for size	Largest struck
class	7	4	3	117
course	10	9	1	225
distance	5	4	1	304
field size	91	64	27	317
going	4	2	2	3

Two sub-families are empty before scoring, and both facts are about the scope of this series rather than the method. Going does not vary: 1,504 of 1,505 search races are “Standard”. Class is too concentrated: classes 2 and 3 hold 117 races between them, under the floor even pooled.

Field size is the one sub-family with a prior. Paper 4 found the overnight forecast-price over-round widening with field size, and the settled starting price does the same.

Table 3: The starting-price over-round by field size, search set. The margin is what a flat backer pays before any skill enters.

Field size	Races	Mean book	Margin
4	45	1.088	8.10%
5	94	1.098	8.96%
6	150	1.117	10.47%
7	154	1.130	11.49%
8	197	1.148	12.87%
9	195	1.158	13.63%
10	171	1.171	14.61%
11	182	1.181	15.36%
12	172	1.200	16.64%
13	98	1.206	17.07%
14	43	1.224	18.33%

The prior is real, and it argues against big fields: a rule that bets them starts further behind.

1.1 Provenance and gates

Nothing here refits anything. The model is paper 5’s rung-3b encoder, scored on the search set by its fitting-partition fit and on the test split by the full-training-split refit paper 5 published. These are two fits of one architecture, so no threshold is assumed to transfer.

Table 4: The two fits.

Fit	Scores	Training races	Con- fig	Epochs	Validation PL loss
fitting partition	validation slice (the search set)	3,517	5	3	5.718496
full training split	test split	5,022	5	3	—

Search set 1,505 races, 13,674 runners; test split 2,183 races, 18,419 runners. Race-id overlap between them, 0; overlap with the 3,517 fitting races, 0. Both asserted in code.

Three gates run first. The ledger reproduces paper 5’s published test backtest to the printed digit, and the search set’s ledger is rebuilt from the source database in code calling none of this paper’s functions, matching on the bet set as well as the arithmetic. The faster sweep used for searching is asserted equal to the ledger on every rule expressible in both. The round-3 scoring path reproduces round 2 on all 25,220 shared rows, maximum absolute ROI difference 0.0e+00.

2 What was found

The rule that won the search returned +37.2% on the search set and -38.1% on the test split. Search-set rank predicts test rank negatively on all four markets. The two best rules on test were the two carried forward without being selected.

Read the mean-price column beside every ROI. Section 3 is why.

2.1 The field-size family

Table 5: Field-size filters, win market, search set, flat stake. Top ten by ROI at the real starting price.

Rule	Bets	Wins	ROI	Fair book	Mean price	Med price	Fav %	Mean field
P4 / field size 11-12	354	42	+37.2%	+65.0%	16.30	13.00	2.3%	11.5
P4 / field size 11-13	452	51	+31.4%	+57.9%	16.82	15.00	2.0%	11.8
P4 / field size 11-14	495	53	+25.2%	+50.4%	17.33	15.00	1.8%	12.0
P4 / field size 11-15	496	53	+24.9%	+50.1%	17.32	15.00	1.8%	12.0
P4 / field size ≥ 11	499	53	+24.2%	+49.2%	17.30	15.00	1.8%	12.0
P4 / field size 10-12	525	52	+16.0%	+38.4%	15.80	13.00	1.7%	11.0
P4 / field size 10-13	623	61	+15.0%	+37.4%	16.25	15.00	1.6%	11.3
P4 / field size 10-14	666	63	+11.5%	+33.2%	16.67	15.00	1.5%	11.5
P4 / field size 10-15	667	63	+11.4%	+33.0%	16.67	15.00	1.5%	11.5
P4 / field size ≥ 10	670	63	+10.9%	+32.4%	16.66	15.00	1.5%	11.5

Table 6: Field-size filters, the other three markets. Top five each.

Market	Rule	Bets	Wins	ROI	Fair book	Mean price	Med price	Fav %	Mean field
Place	P4 / field size 11-12	354	98	+9.0%	+30.9%	16.30	13.00	2.3%	11.5
Place	P4 / field size 11-13	452	121	+8.8%	+30.8%	16.82	15.00	2.0%	11.8
Place	P4 / field size 11-14	495	129	+6.1%	+27.5%	17.33	15.00	1.8%	12.0
Place	P4 / field size 11-15	496	129	+5.9%	+27.2%	17.32	15.00	1.8%	12.0
Place	P4 / field size ≥ 11	499	129	+5.2%	+26.5%	17.30	15.00	1.8%	12.0

Table 6: Field-size filters, the other three markets. Top five each.

Market	Rule	Bets	Wins	ROI	Fair book	Mean price	Med price	Fav %	Mean field
Each-way (paper 5 terms)	P4 / field size 11-12	354	98	+15.8%	+37.0%	16.30	13.00	2.3%	11.5
Each-way (paper 5 terms)	P4 / field size 11-13	452	121	+12.3%	+32.8%	16.82	15.00	2.0%	11.8
Each-way (paper 5 terms)	P4 / field size 11-14	495	129	+7.9%	+27.6%	17.33	15.00	1.8%	12.0
Each-way (paper 5 terms)	P4 / field size 11-15	496	129	+7.7%	+27.3%	17.32	15.00	1.8%	12.0
Each-way (paper 5 terms)	P4 / field size ≥ 11	499	129	+7.0%	+26.5%	17.30	15.00	1.8%	12.0
Each-way (corrected terms)	P4 / field size 11-12	354	98	+20.6%	+42.8%	16.30	13.00	2.3%	11.5
Each-way (corrected terms)	P4 / field size 11-13	452	121	+17.7%	+39.4%	16.82	15.00	2.0%	11.8
Each-way (corrected terms)	P4 / field size 11-14	495	129	+13.4%	+34.2%	17.33	15.00	1.8%	12.0
Each-way (corrected terms)	P4 / field size 11-15	496	129	+13.2%	+34.0%	17.32	15.00	1.8%	12.0
Each-way (corrected terms)	P4 / field size ≥ 11	499	129	+12.5%	+33.2%	17.30	15.00	1.8%	12.0

One rule tops every market: P4 in fields of 11 to 12 runners. Back the horse whose model probability most exceeds its market probability, in a middling-to-large field. Mean backed price 16.30.

2.2 Other sub-families, and the combined layer

Table 7: The race-characteristic family pooled across sub-families, search set. Top six per market.

Market	Rule	Bets	Wins	ROI	Fair book	Mean price	Med price	Fav %	Mean field
Win	P4 / field size 11-12	354	42	+37.2%	+65.0%	16.30	13.00	2.3%	11.5
Win	P4 / field size 11-13	452	51	+31.4%	+57.9%	16.82	15.00	2.0%	11.8
Win	P4 / field size 11-14	495	53	+25.2%	+50.4%	17.33	15.00	1.8%	12.0
Win	P4 / field size 11-15	496	53	+24.9%	+50.1%	17.32	15.00	1.8%	12.0
Win	P4 / field size >= 11	499	53	+24.2%	+49.2%	17.30	15.00	1.8%	12.0
Win	P4 / course Wolverhampton only	453	51	+18.8%	+40.4%	13.79	12.00	4.0%	9.2
Place	P4 / field size 11-12	354	98	+9.0%	+30.9%	16.30	13.00	2.3%	11.5
Place	P4 / field size 11-13	452	121	+8.8%	+30.8%	16.82	15.00	2.0%	11.8
Place	P4 / field size 11-14	495	129	+6.1%	+27.5%	17.33	15.00	1.8%	12.0
Place	P4 / field size 11-15	496	129	+5.9%	+27.2%	17.32	15.00	1.8%	12.0
Place	P4 / field size >= 11	499	129	+5.2%	+26.5%	17.30	15.00	1.8%	12.0
Place	P4 / course Wolverhampton only	453	157	+4.6%	+22.7%	13.79	12.00	4.0%	9.2
Each-way (paper 5 terms)	P4 / field size 11-12	354	98	+15.8%	+37.0%	16.30	13.00	2.3%	11.5
Each-way (paper 5 terms)	P4 / field size 11-13	452	121	+12.3%	+32.8%	16.82	15.00	2.0%	11.8
Each-way (paper 5 terms)	P4 / course Wolverhampton only	453	157	+9.5%	+26.5%	13.79	12.00	4.0%	9.2

Table 7: The race-characteristic family pooled across sub-families, search set. Top six per market.

Market	Rule	Bets	Wins	ROI	Fair book	Mean price	Med price	Fav %	Mean field
Each-way (paper 5 terms)	P4 / field size 11-14	495	129	+7.9%	+27.6%	17.33	15.00	1.8%	12.0
Each-way (paper 5 terms)	P4 / field size 11-15	496	129	+7.7%	+27.3%	17.32	15.00	1.8%	12.0
Each-way (paper 5 terms)	P4 / field size ≥ 11	499	129	+7.0%	+26.5%	17.30	15.00	1.8%	12.0
Each-way (corrected terms)	P4 / field size 11-12	354	98	+20.6%	+42.8%	16.30	13.00	2.3%	11.5
Each-way (corrected terms)	P4 / field size 11-13	452	121	+17.7%	+39.4%	16.82	15.00	2.0%	11.8
Each-way (corrected terms)	P4 / field size 11-14	495	129	+13.4%	+34.2%	17.33	15.00	1.8%	12.0
Each-way (corrected terms)	P4 / field size 11-15	496	129	+13.2%	+34.0%	17.32	15.00	1.8%	12.0
Each-way (corrected terms)	P4 / field size ≥ 11	499	129	+12.5%	+33.2%	17.30	15.00	1.8%	12.0
Each-way (corrected terms)	P4 / course Wolverhampton only	453	128	+6.6%	+24.0%	13.79	12.00	4.0%	9.2

Course and distance contribute a little. Class and going contribute nothing. Field size does the work.

Table 8: The combined layer, which crosses the best race-characteristic filter per picker and market with the full threshold grid, against the pure race-characteristic rule.

Market	Best race-characteristic rule	Its ROI	Best combined rule	Com- bined ROI	Gain
Win	P4 / field size 11-12	+37.2%	P4 / field size 11-12 + P>0.11, ratio>1.05	+46.4%	+9.2%
Place	P4 / field size 11-12	+9.0%	P4 / field size 11-12 + P>0.07, ratio>1.50	+13.3%	+4.3%
Each-way (paper 5 terms)	P4 / field size 11-12	+15.8%	P4 / field size 11-12 + P>0.11, ratio>1.05	+19.2%	+3.3%
Each-way (corrected terms)	P4 / field size 11-12	+20.6%	P4 / field size 11-12 + P>0.07, ratio>1.30	+23.7%	+3.1%

Adding the model-probability plane gains a few points. Every winning combined rule is the same field-size band with a light cut on it.

2.3 The stability guard

The window splits at 2012-01-15 into 755 and 750 races. A candidate reaches test only if it ranks in the top decile of its family on both halves with at least 150 bets in each.

Table 9: What the guard strikes, by family and market.

Family	Bet	Candi- dates	Eligi- ble	Struck: too few bets	Struck: not top decile	Sur- vivors
combined	Each-way (corrected terms)	11349	1695	0	11182	167
combined	Each-way (paper 5 terms)	11349	1695	0	11203	146
combined	Place	11349	1695	0	11346	3
combined	Win	11349	1695	0	11191	158
race charac- teristic	Each-way (corrected terms)	415	415	0	401	14
race charac- teristic	Each-way (paper 5 terms)	415	415	0	403	12
race charac- teristic	Place	415	415	0	396	19
race charac- teristic	Win	415	415	0	400	15

Table 9: What the guard strikes, by family and market.

Family	Bet	Candi- dates	Eligi- ble	Struck: too few bets	Struck: not top decile	Sur- vivors
threshold	Each-way (corrected terms)	6305	2880	0	6257	48
threshold	Each-way (paper 5 terms)	6305	2880	0	6275	30
threshold	Place	6305	2880	0	6248	57
threshold	Win	6305	2880	0	6276	29

Owen’s rule fails the guard on three of the four markets. The rule this series has used since paper 1 is not stable inside the search window.

Table 10: Owen’s rule against the guard.

Market	ROI	H1 ROI	H2 ROI	H1 rank	H2 rank	of	Cutoff	Sur- vives
Win	+8.0%	+22.6%	- 10.2%	8	871	2880	288	no
Place	-3.8%	+0.6%	-9.4%	217	288	2880	288	yes
Each-way (paper 5 terms)	+5.5%	+15.2%	-6.7%	10	773	2880	288	no
Each-way (corrected terms)	-3.1%	+5.2%	- 13.4%	11	1264	2880	288	no

The cross-half rank correlation says whether searching a family on 1,505 races can work at all.

Table 11: Cross-half rank correlation of ROI within each family.

Family	Market	Candi- dates	Spear- man	Pearson	Survivors
combined	Each-way (corrected terms)	1695	+0.875	+0.938	167
combined	Each-way (paper 5 terms)	1695	+0.787	+0.882	146
combined	Place	1695	-0.139	+0.024	3
combined	Win	1695	+0.565	+0.893	158
race characteristic	Each-way (corrected terms)	415	+0.592	+0.772	14

Table 11: Cross-half rank correlation of ROI within each family.

Family	Market	Candi- dates	Spear- man	Pearson	Survivors
race characteristic	Each-way (paper 5 terms)	415	+0.514	+0.685	12
race characteristic	Place	415	+0.329	+0.401	19
race characteristic	Win	415	+0.530	+0.671	15
threshold	Each-way (corrected terms)	2880	+0.585	+0.732	48
threshold	Each-way (paper 5 terms)	2880	+0.420	+0.615	30
threshold	Place	2880	+0.245	+0.300	57
threshold	Win	2880	+0.432	+0.556	29

Table 12: The same, within the race-characteristic sub-families.

Sub-family	Market	Candidates	Spearman	Survivors
class	Each-way (corrected terms)	20	+0.689	0
class	Each-way (paper 5 terms)	20	+0.535	0
class	Place	20	+0.252	0
class	Win	20	+0.621	0
course	Each-way (corrected terms)	45	+0.357	1
course	Each-way (paper 5 terms)	45	+0.434	3
course	Place	45	+0.418	3
course	Win	45	+0.362	1
distance	Each-way (corrected terms)	20	+0.745	1
distance	Each-way (paper 5 terms)	20	+0.638	1
distance	Place	20	+0.567	1
distance	Win	20	+0.611	1
field size	Each-way (corrected terms)	320	+0.612	12
field size	Each-way (paper 5 terms)	320	+0.532	8
field size	Place	320	+0.322	15
field size	Win	320	+0.565	13
going	Each-way (corrected terms)	10	+0.890	0
going	Each-way (paper 5 terms)	10	+0.399	0
going	Place	10	+0.877	0
going	Win	10	+0.313	0

Nothing reaches a level at which one window is a reliable guide. Place is the least consistent everywhere. The combined layer’s high correlations are an artefact: every combined candidate

contains the same race-characteristic filter, so they are near-copies and their halves agree because they are barely different rules.

2.4 The test split, scored once

Carried forward per market: the top three surviving race-characteristic candidates, the top two surviving combined candidates, Owen’s rule, and the market favourite in every race. The last two were carried forward unconditionally, as comparator and control.

Table 13: Win market: search set to test split, ordered by search-set ROI.

Rule	Fam- ily	Val ROI	Test				Test less		Val bets	Test bets	Val price	Test price	Val rank	Test rank
			H1	H2	ROI	CI	90%	Test top fair win						
P4 / field size 11-12 + P>0.10, ratio>1.05	com- bined	+43.3%	43.3%	43.3%	31.6%	[-31.6%, 16.9%]	58.0%	-	339	323	15.70	18.26	1	5
P4 / field size 11-12 + P>0.10, ratio>1.10	com- bined	+42.7%	38.1%	46.5%	29.2%	[-29.2%, +0.9%]	56.3%	-	330	312	15.49	17.78	2	4
P4 / field size 11-12	race char- acter- istic	+37.2%	38.0%	36.6%	38.1%	[-38.1%, 12.3%]	62.4%	-	354	357	16.30	18.88	3	7
P4 / field size 11-13	race char- acter- istic	+31.4%	22.6%	38.8%	35.2%	[-35.2%, 12.2%]	57.1%	-	452	432	16.82	18.83	4	6
P4 / field size 11-14	race char- acter- istic	+25.2%	21.6%	28.3%	26.1%	[-26.1%, 1.1%]	50.9%	-	495	471	17.33	18.74	5	3
P5 / P>0.15, ratio>1.30	thresh- old	+8.0%	22.6%	-	10.2%	[-10.2%, +0.6%]	24.8%	+0.8%	704	1064	7.71	8.02	6	2
P2 / no filter	thresh- old	-	-	-	-	[-13.1%, 1.5%]	11.3%	+8.4%	1505	2183	3.48	3.15	7	1

Table 14: Place market: search set to test split.

Rule	Fam- ily	Val		Test		Test		Test less top win	Val bets	Test bets	Val price	Test price	Val rank	Test rank
		ROI	H1 H2	ROI	CI	90%	Test							
P4 / field size 11-12 + P>0.05, ratio>1.50	com- bined	+10.4%	3.6% 16.0%	29.1%	[-	-	-	-	332	349	16.61	19.03	1	6
P4 / field size 11-12 + P>0.07, ratio>1.40	com- bined	+10.3%	5.2% 14.6%	27.2%	[-	-	-	-	327	338	15.94	18.29	2	5
P4 / field size 11-12	race char- acter- istic	+9.0%	+17.1% 0.5%	29.5%	[-	-	-	-	354	357	16.30	18.88	3	7
P4 / field size 11-13	race char- acter- istic	+8.8%	6.1% 11.2%	24.2%	[-	-	-	-	452	432	16.82	18.83	4	4
P4 / field size 11-14	race char- acter- istic	+6.1%	5.1% 6.9%	22.2%	[-	-	-	-	495	471	17.33	18.74	5	3
P5 / P>0.15, ratio>1.30	thresh- old	-	+0.6% 3.8%	-	[-	-	-	-	704	1064	7.71	8.02	6	1
P2 / no filter	thresh- old	-	-	-	[-	-	-	-	1505	2183	3.48	3.15	7	2

Table 15: Each-way at paper 5’s flat one-fifth, top-three terms.

Rule	Fam- ily	Val		Test		Test		Test less top win	Val bets	Test bets	Val price	Test price	Val rank	Test rank
		ROI	H1 H2	ROI	CI	90%	Test							
P4 / field size 11-12 + P>0.07, ratio>1.30	com- bined	+18.8%	14.8% 22.1%	37.2%	[-	-	-	-	343	349	15.87	18.39	1	6
						53.5%, 25.4%	41.6%							
						-	20.4%]							
P4 / field size 11-12 + P>0.11, ratio>0.95	com- bined	+18.5%	12.0% 24.0%	33.3%	[-	-	-	-	329	313	16.01	18.20	2	4
						51.4%, 20.7%	38.3%							
						-	14.4%]							
P4 / field size 11-12	race char- acter- istic	+15.8%	12.0% 19.1%	38.6%	[-	-	-	-	354	357	16.30	18.88	3	7
						54.4%, 27.1%	42.9%							
						-	21.9%]							
P4 / field size 11-13	race char- acter- istic	+12.3%	6.3% 17.3%	35.2%	[-	-	-	-	452	432	16.82	18.83	4	5
						49.6%, 22.7%	38.7%							
						-	19.8%]							
P4 / course Wolver- hampton only	race char- acter- istic	+9.5%	6.0% 12.5%	0.3%	[-		+12.4%		453	658	13.79	14.94	5	1
						14.7%, 2.9%								
						+14.3%]								
P5 / P>0.15, ratio>1.30	thresh- old	+5.5%	15.2% -	6.7%	[-		+5.1%		704	1064	7.71	8.02	6	2
						6.0% 14.0%,	6.8%							
						+2.0%]								
P2 / no filter	thresh- old	-	- -	-	[-		+3.8%		1505	2183	3.48	3.15	7	3
						10.6% 12.2% 8.9% 6.7%	10.1%,							
						-	3.5%]							

Table 16: Each-way at the corrected industry ladder. These are the terms a bet settles on.

Rule	Fam- ily	Val		Test		Test		Test		Val bets	Test bets	Val price	Test price	Val rank	Test rank
		ROI	H1 H2	ROI	90% CI	Test	90% CI	Test	less top						
P4 / field size 11-12 + P>0.07, ratio>1.30	com- bined	+23.7%	18.5% 28.0%	34.5%	[-11.2%, 22.1%]	-	-	-	-	343	349	15.87	18.39	1	6
P4 / field size 11-12 + P>0.11, ratio>0.95	com- bined	+23.1%	15.3% 29.8%	30.8%	[-49.4%, 17.5%]	-	-	-	-	329	313	16.01	18.20	2	4
P4 / field size 11-12	race char- acter- istic	+20.6%	15.5% 24.8%	36.0%	[-52.3%, 23.8%]	-	-	-	-	354	357	16.30	18.88	3	7
P4 / field size 11-13	race char- acter- istic	+17.7%	11.3% 23.3%	31.6%	[-46.8%, 18.4%]	-	-	-	-	452	432	16.82	18.83	4	5
P4 / field size 11-14	race char- acter- istic	+13.4%	10.4% 16.0%	26.2%	[-42.3%, 11.4%]	-	-	-	-	495	471	17.33	18.74	5	3
P5 / P>0.15, ratio>1.30	thresh- old	- 3.1%	+5.2% 13.4%	- 17.4%	[-25.7%, 6.9%]	-	-	-	-	704	1064	7.71	8.02	6	2
P2 / no filter	thresh- old	- 11.7%	- 13.8% 0.7%	- 8.8%	[-12.3%, 5.4%]	-	+2.1%	-	-	1505	2183	3.48	3.15	7	1

The search winner reversed by 75 points. It returned +38.0% and +36.6% on the two halves of the window, then -38.1% on test, 90% interval [-62.4%, -12.3%]. The best combined rule went from +43.3% to -31.6%.

The price column predicted it and the guard did not. Mean backed price rose from 16.30 to 18.88 and the win strike rate fell from 11.9% to 5.3%. Maximising the gap between model

and market probability in a field big enough to hold a long price means backing the horse the model is most wrong about. It paid on one window.

It also loses at a zero-margin book, returning -24.8%. A rule that only loses the bookmaker's margin is a candidate for a better price. A rule that loses with the margin removed is wrong.

The market favourite in every race, with no model input, returned -6.4%; Owen's rule returned -12.2%. On the search set they placed last and sixth of seven.

Table 17: Every rule profitable at a zero-margin book on the test split: three of the 28 rule-by-market combinations scored, and no search chose any of them.

Market	Rule	Bets	Test ROI	Test fair book	Mean price	90% CI
Each-way (paper 5 terms)	P4 / course	658	-0.3%	+12.4%	14.94	[-14.7%, +14.3%]
Win	Wolverhampton only					
	P2 / no filter	2183	-6.4%	+8.4%	3.15	[-11.3%, -1.5%]
Each-way (paper 5 terms)	P5 / P>0.15, ratio>1.30	1064	-6.0%	+5.1%	8.02	[-14.0%, +2.0%]
Each-way (paper 5 terms)	P2 / no filter	2183	-6.7%	+3.8%	3.15	[-10.1%, -3.5%]
Place	P5 / P>0.15, ratio>1.30	1064	-10.2%	+2.7%	8.02	[-15.7%, -4.6%]
Each-way (corrected terms)	P2 / no filter	2183	-8.8%	+2.1%	3.15	[-12.3%, -5.4%]
Win	P5 / P>0.15, ratio>1.30	1064	-12.2%	+0.8%	8.02	[-24.8%, +0.6%]

Table 18: Search-set rank against test rank, within each market's shortlist.

Market	Rules	Spearman	Pearson
Each-way (paper 5 terms)	7	-0.679	-0.713
Each-way (corrected terms)	7	-0.821	-0.974
Place	7	-0.857	-0.683
Win	7	-0.714	-0.912

Table 19: The same, pooled by family across markets.

Family	Rules	Spearman	Pearson	Mean search ROI	Mean test ROI
combined	8	-0.381	-0.031	+23.8%	-31.6%
race characteristic	12	-0.657	-0.510	+17.3%	-28.6%
threshold	8	+0.095	+0.150	-5.8%	-10.8%

Search-set ROI selects nothing useful. The correlation is negative on all four markets and in both families the search produced.

Round 2 reported a Spearman of +0.857 on the place market and read it as evidence that a held-out window selects usefully there. It did not replicate: with a different shortlist on the same market the correlation is -0.857. Seven-rule rank correlations are themselves noisy. A 1,505-race window does not rank betting rules.

3 Why most of the surface is price band

Two thirds to four fifths of the difference in return between rules is explained by the mean price the rule backs. A leaderboard on this data mostly ranks price bands.

Table 20: ROI against mean backed price, across the threshold family on the search set.

Market	Rules	cor(ROI, price)	cor(ROI, log price)	R^2 , quadratic in log price	Residual SD	Fitted peak at price	Median rule ROI
Win	2990	-0.798	-0.725	68.8%	7.5%	5.60	-11.7%
Place	2990	-0.396	-0.187	64.6%	3.6%	8.52	-14.6%
Each-way (paper 5 terms)	2990	-0.872	-0.778	83.0%	4.5%	5.84	-10.0%
Each-way (corrected terms)	2990	-0.881	-0.823	79.8%	5.4%	4.80	-13.1%

The surface is a map of the favourite-longshot bias read at different prices, with an interior optimum where the tilt to longer prices stops paying for the extra variance and margin.

That is why P4 in fields of 11 to 12 topped every search-set table. Its mean backed price is 16.30, against 4.76 for the model's top-rated horse and 15.82 for the average runner in an 11-or-12-horse field. The second comparison is the informative one. P4 in a big field backs a

horse priced like a runner drawn at random from it, and only 2.3% of its picks are the favourite. It is not selecting a price band. It is declining to select, in the races where declining costs most.

Table 21: Owen’s rule against the price-band fit. The residual is the part of its return price does not explain.

Market	Owen ROI	Mean price	Fitted from price alone	Residual	Residual in SDs	Rules ranked	Owen’s rank
Win	+8.0%	7.71	-6.9%	+14.9%	+1.99	2990	13
Place	-3.8%	7.71	-7.7%	+3.9%	+1.07	2990	131
Each-way (paper 5 terms)	+5.5%	7.71	-5.1%	+10.6%	+2.33	2990	9
Each-way (corrected terms)	-3.1%	7.71	-10.9%	+7.9%	+1.45	2990	59

Owen’s win-market residual is 1.99 residual standard deviations. That is not a finding on its own, because the residual does not survive halving the window.

Split at 2012-01-15, Owen’s cell moves from +22.6% to -10.2%, and from rank 8 to rank 871 of 2,880. An earlier sweep of Owen’s two thresholds alone put the same cell first of 617 on the first half and 157th of 579 on the second, while also placing it second of 597 rules over the full window. That is what a peak looks like when a two-year window rewards one price band: the 6-to-9 band in that window returned about ten points more against its implied probability than the same band does on test.

So a leaderboard on 1,505 races mostly ranks price bands, and the part that is not price band does not hold across halves of one window.

The guard in Section 2 was built for that, and it struck Owen’s rule on three of four markets. It still did not save the search: the rule it passed returned +38.0% and +36.6% on the two halves and -38.1% outside the window. Within-window stability is a different property from across-window transfer. This data has the first and not the second.

4 Staking

Neither non-flat staking rule helps. Of the 32 non-flat arms scored across the two splits, 29 return less than their own flat-stake counterpart.

Every search in this paper runs at a flat one-unit stake, and staking is varied separately on a fixed selection. In the first attempt at this paper the non-flat rules were searched alongside the selection rules. Because they stake zero wherever the model’s probability does not exceed the

price, they changed the bet set as well as the stake, and the search preferred them for a reason that had nothing to do with staking.

Three rules: flat at one unit; edge-proportional, $(\hat{P} - P^{\text{mkt}})$ over a five-point edge, capped at three units; and quarter-Kelly at $\hat{P}$ against the starting price, on the same cap.

Table 22: Staking on the test split, selection held fixed. The zero-stake share is the fraction of selected races where the staking rule puts nothing on.

Market	Rule	Staking	Races selected	Races staked	Zero-stake share	Total staked	ROI	Fair book
Win	P3 / $P > 0.15$, ratio > 1.70	K0	534	534	0.0%	534.0	-6.0%	+7.5%
Win	P3 / $P > 0.15$, ratio > 1.70	K1	534	534	0.0%	927.9	-16.2%	-4.4%
Win	P3 / $P > 0.15$, ratio > 1.70	K2	534	534	0.0%	231.8	-18.7%	-7.4%
Win	P5 / $P > 0.15$, ratio > 1.30	K0	1064	1064	0.0%	1064.0	-12.2%	+0.8%
Win	P5 / $P > 0.15$, ratio > 1.30	K1	1064	1064	0.0%	1759.8	-15.8%	-3.5%
Win	P5 / $P > 0.15$, ratio > 1.30	K2	1064	1064	0.0%	407.6	-18.2%	-6.6%
Place	P3 / $P > 0.15$, ratio > 1.70	K0	534	534	0.0%	534.0	+0.6%	+14.9%
Place	P3 / $P > 0.15$, ratio > 1.70	K1	534	534	0.0%	927.9	-3.9%	+9.2%
Place	P3 / $P > 0.15$, ratio > 1.70	K2	534	534	0.0%	231.8	-3.7%	+9.2%
Place	P5 / $P > 0.15$, ratio > 1.30	K0	1064	1064	0.0%	1064.0	-10.2%	+2.7%
Place	P5 / $P > 0.15$, ratio > 1.30	K1	1064	1064	0.0%	1759.8	-10.7%	+1.9%
Place	P5 / $P > 0.15$, ratio > 1.30	K2	1064	1064	0.0%	407.6	-9.8%	+2.2%
Each-way (paper 5 terms)	P5 / $P > 0.15$, ratio > 1.30	K0	1064	1064	0.0%	2128.0	-6.0%	+5.1%
Each-way (paper 5 terms)	P5 / $P > 0.15$, ratio > 1.30	K1	1064	1064	0.0%	3519.6	-7.2%	+3.4%

Table 22: Staking on the test split, selection held fixed. The zero-stake share is the fraction of selected races where the staking rule puts nothing on.

Market	Rule	Stak- ing	Races selected	Races staked	Zero- stake share	Total staked	ROI	Fair book
Each-way (paper 5 terms)	P5 / P>0.15, ratio>1.30	K2	1064	1064	0.0%	815.1	- 7.3%	+2.9%
Each-way (paper 5 terms)	P5 / P>0.17, ratio>1.45	K0	589	589	0.0%	1178.0	- 0.3%	+10.9%
Each-way (paper 5 terms)	P5 / P>0.17, ratio>1.45	K1	589	589	0.0%	2331.7	- 3.8%	+6.8%
Each-way (paper 5 terms)	P5 / P>0.17, ratio>1.45	K2	589	589	0.0%	574.7	- 4.5%	+5.6%
Each-way (corrected terms)	P2 / P>0.10, ratio>2.25	K0	554	554	0.0%	1084.0	- 6.3%	+4.2%
Each-way (corrected terms)	P2 / P>0.10, ratio>2.25	K1	554	13	97.7%	13.0	- 37.9%	- 31.9%
Each-way (corrected terms)	P2 / P>0.10, ratio>2.25	K2	554	6	98.9%	0.4	- 57.1%	- 53.7%
Each-way (corrected terms)	P5 / P>0.15, ratio>1.30	K0	1064	1064	0.0%	2063.0	- 17.4%	-6.9%
Each-way (corrected terms)	P5 / P>0.15, ratio>1.30	K1	1064	1064	0.0%	3379.7	- 20.1%	- 10.0%
Each-way (corrected terms)	P5 / P>0.15, ratio>1.30	K2	1064	1064	0.0%	775.8	- 21.6%	- 12.0%

Table 23: The same on the search set.

Market	Rule	Stak- ing	Races selected	Races staked	Zero- stake share	Total staked	ROI	Fair book
Win	P3 / $P > 0.15$, ratio > 1.70	K0	328	328	0.0%	328.0	+10.0%	+26.2%
Win	P3 / $P > 0.15$, ratio > 1.70	K1	328	328	0.0%	598.8	- 6.8%	+6.0%
Win	P3 / $P > 0.15$, ratio > 1.70	K2	328	328	0.0%	154.8	- 6.2%	+6.3%
Win	P5 / $P > 0.15$, ratio > 1.30	K0	704	704	0.0%	704.0	+8.0%	+23.4%
Win	P5 / $P > 0.15$, ratio > 1.30	K1	704	704	0.0%	1163.4	+3.9%	+18.3%
Win	P5 / $P > 0.15$, ratio > 1.30	K2	704	704	0.0%	275.3	+2.7%	+16.4%
Place	P3 / $P > 0.15$, ratio > 1.70	K0	328	328	0.0%	328.0	- 0.1%	+14.3%
Place	P3 / $P > 0.15$, ratio > 1.70	K1	328	328	0.0%	598.8	- 8.9%	+3.7%
Place	P3 / $P > 0.15$, ratio > 1.70	K2	328	328	0.0%	154.8	- 11.4%	+0.6%
Place	P5 / $P > 0.15$, ratio > 1.30	K0	704	704	0.0%	704.0	- 3.8%	+10.1%
Place	P5 / $P > 0.15$, ratio > 1.30	K1	704	704	0.0%	1163.4	- 6.1%	+7.2%
Place	P5 / $P > 0.15$, ratio > 1.30	K2	704	704	0.0%	275.3	- 8.2%	+4.3%
Each-way (paper 5 terms)	P5 / $P > 0.15$, ratio > 1.30	K0	704	704	0.0%	1408.0	+5.5%	+17.9%
Each-way (paper 5 terms)	P5 / $P > 0.15$, ratio > 1.30	K1	704	704	0.0%	2326.9	+2.9%	+14.7%
Each-way (paper 5 terms)	P5 / $P > 0.15$, ratio > 1.30	K2	704	704	0.0%	550.7	+1.7%	+12.9%
Each-way (paper 5 terms)	P5 / $P > 0.17$, ratio > 1.45	K0	370	370	0.0%	740.0	+6.8%	+18.9%

Table 23: The same on the search set.

Market	Rule	Staking	Races selected	Races staked	Zero-stake share	Total staked	ROI	Fair book
Each-way (paper 5 terms)	P5 / $P > 0.17$, ratio > 1.45	K1	370	370	0.0%	1481.7	+3.5%	+14.9%
Each-way (paper 5 terms)	P5 / $P > 0.17$, ratio > 1.45	K2	370	370	0.0%	375.1	+2.0%	+12.8%
Each-way (corrected terms)	P2 / $P > 0.10$, ratio > 2.25	K0	311	311	0.0%	612.0	+3.4%	+14.9%
Each-way (corrected terms)	P2 / $P > 0.10$, ratio > 2.25	K1	311	14	95.5%	9.5	+40.2%	+56.7%
Each-way (corrected terms)	P2 / $P > 0.10$, ratio > 2.25	K2	311	2	99.4%	0.6	+88.4%	+103.9%
Each-way (corrected terms)	P5 / $P > 0.15$, ratio > 1.30	K0	704	704	0.0%	1382.0	- 3.1%	+9.0%
Each-way (corrected terms)	P5 / $P > 0.15$, ratio > 1.30	K1	704	704	0.0%	2271.9	- 6.6%	+4.9%
Each-way (corrected terms)	P5 / $P > 0.15$, ratio > 1.30	K2	704	704	0.0%	534.7	- 8.4%	+2.5%

Two mechanisms. Where the selection already requires the model's probability to exceed the price, the zero-stake share is 0.0% and the staking rules lever up an identical bet set, so a negative return becomes more negative. Where it does not, the stake collapses.

Table 24: The arms where the staking rule discards most of the bet set. An ROI on this many units is not a staking result.

Split	Market	Rule	Staking	Races selected	Races staked	Zero-stake share	Total staked	ROI
test	Each-way (corrected terms)	P2 / $P > 0.10$, ratio > 2.25	K1	554	13	97.7%	13.0	-37.9%
test	Each-way (corrected terms)	P2 / $P > 0.10$, ratio > 2.25	K2	554	6	98.9%	0.4	-57.1%
search set	Each-way (corrected terms)	P2 / $P > 0.10$, ratio > 2.25	K1	311	14	95.5%	9.5	+40.2%
search set	Each-way (corrected terms)	P2 / $P > 0.10$, ratio > 2.25	K2	311	2	99.4%	0.6	+88.4%

That is the market-favourite rules: the model usually rates the favourite below the market, so both non-flat rules stake nothing on most selected races. At worst the rule stakes 2 of 554. Report the zero-stake share, not the ROI alone. In the first attempt at this paper the declared win rule staked 33 units across 2,183 test races with an interval 54 points wide, and none of that was visible in its ROI.

5 How the model will likely be used

This section is informed speculation. Nothing in it is a result of this paper.

5.1 Every number in this series is measured at the wrong price

Paper 4 asked whether the model adds anything given a market price. Two answers came back.

Against the **settled starting price** it adds nothing that can be distinguished from zero. The pooled model-probability coefficient in a two-stage conditional logit was 0.068 for the paper-2b ranking model and 0.039 for the gradient-boosted trees over 2,182 races, with 95% intervals reaching no higher than 0.204.

Against the **overnight racecard forecast price** it adds a great deal: 1.037 and 1.026 on the earlier test half, 0.746 and 0.723 on the later.

So the model knows something the overnight price does not and nothing the settled price does not. Its value sits in the hours between those two moments. Every backtest in papers 1 to 6 settles at the closing end of that interval, which is the one price where the model was already known to add nothing.

That reframes Section 2. The searches were not failing to find a good rule. They were looking for a rule strong enough to beat a margin at a price where the model has no informational advantage left.

5.2 The likely use is a price trade, not a winner-picking trade

Backing at the off means picking winners against a market that has already absorbed everything the model knows. Backing early means taking a price the model thinks is too long, which is a different bet with a different payoff.

The trade has two exits. Hold to the result and collect at the early price if the horse wins. Or lay the same horse back on the exchange once the price shortens, taking the difference and carrying no result risk at all. The second exit is available whether or not the horse runs well, which is the point: the bet is on the price moving, not on the horse.

The mechanism this leans on is convergence. Early markets are thin and dominated by bookmaker prices and money that has not looked closely. As race time approaches, liquidity builds and prices move toward fair value. A model that approximates the closing price from pre-race information alone identifies horses whose early price has not yet moved. The edge, if it exists, is in front-running that convergence.

Paper 4's numbers are consistent with that and do not establish it. What they establish is where to look.

5.3 The metric changes with the trade

Judge it by closing line value, not ROI. Did the price taken beat the price at the off?

That is a per-bet measurement rather than a per-win one. It is available on every bet, it does not wait for a result, and it converges in hundreds of bets rather than tens of thousands. ROI on a win market converges slowly because almost all its variance sits in whether a few long-priced bets landed. The field-size rule's entire 75-point reversal in Section 2 is 42 winners becoming 19.

This is the reason every ROI interval in this series is too wide to read. The intervals in Section 2 run 20 to 50 points wide on samples of 300 to 1,000 bets. A metric that resolves in hundreds of bets is not a refinement here. It is the difference between being able to tell whether the strategy works and not being able to tell.

If the convergence thesis is right, closing line value is positive and shows up quickly. If it is wrong, that shows up quickly too, and long before an ROI would say anything.

5.4 The exchange changes the arithmetic

Bookmaker prices carry the over-round in Section 1: from 8.1% at four runners to 18.3% at fourteen. Field sizes 8 to 12 hold 61% of the search races and pay between 12.9% and 16.6%.

An exchange charges commission on net winnings instead, typically 2 to 5%. That gap is larger than any effect this paper searched for. The best test-split return in the whole shortlist is -0.3%, and the spread between the best and worst rules is smaller than the difference between paying an 8-to-18% over-round and paying 2-to-5% commission.

The fair-book column makes the same point from the other side. Three rules in the shortlist are profitable with the margin removed and none is profitable with it in place. Changing the venue does not create an edge. It changes how much edge a rule needs before one survives.

5.5 Two things this dataset cannot answer

Is real money available at early prices, in usable size? Everything above assumes the bet can be struck. The Smartform archive records what prices existed, not what money stood behind them. Early exchange markets on UK All-Weather handicaps are thin. A price available for one unit may not be available for fifty, and a price that moves when you bet into it is not the price you modelled. If the available size is small enough, the edge is real and unusable, which is the same thing operationally as not having it.

Does the model beat an early exchange price? Paper 4's result is against `daily_runners.forecast_price_decimal`, a forecast published the evening before by a compiler. That is not a price money has traded at. An early exchange price aggregates the opinions of people who have committed funds, and it will already contain much of what a compiler's forecast misses. Beating a forecast price is weak evidence of beating an early traded price. This dataset contains no early traded prices, so the gap cannot be measured here.

Neither question is a footnote. They decide whether any of this works, and answering them needs a live feed and paper trading, not another backtest on this archive.

5.6 A correction to the series

Two of the four markets reported throughout papers 2b to 6 price bets that do not quite exist on the venue this strategy would use.

Betfair has no each-way market. An each-way bet is a bookmaker product: one stake on the win, one on a place at a fraction of the odds. On an exchange those are two separate

markets and the each-way construction has no counterpart. Every each-way figure in papers 2b, 5 and 6, at paper 5's flat one-fifth top-three terms and at the corrected industry ladder alike, prices something that cannot be placed there.

Betfair's place market is independently priced. The place prices in this series are not. They are a discounted-Harville construction computed from the win prices, following Harville (1973) with the Lo and Bacon-Shone (1994) corrections. That was the right choice for papers 2b to 5, because it makes the place market comparable to the win market on the same information. It is not the price a place bet would be struck at, and a real place market can disagree with anything derived from win prices.

Of the four markets in this paper, win is real and priced as traded, place is real but priced here by proxy, and the two each-way columns are products of a different venue. The win numbers are the ones that carry forward.

6 Limitations

The search window is too small to rank rules. It is 1,505 races. Cross-half rank correlations run between -0.139 and +0.875, and the rule that won on both halves reversed by 75 points out of sample. Nothing here has identified a good rule.

Search-set rank predicts test rank negatively on all four markets. Round 2's positive place-market correlation did not replicate. The only difference between the two rounds is which seven rules were in the shortlist.

Twenty-eight rule-by-market combinations were read against one test split with no multiplicity correction. The intervals are marginal 90% race-level bootstraps and are not adjusted for having looked twenty-eight times. Whatever tops a test table is partly noise. The two rules that did top them were carried forward unconditionally rather than selected, which is the only reason their position means anything.

Only the betting rule is clean. The model was selected with knowledge of these test races across five papers: features, objective, function class, encoder architecture and the rung ladder were each chosen after seeing earlier papers' test results. The split has been scored once per paper, six times on the same races. Paper 6 fits nothing, so its betting rules are out of sample. The model underneath them is not.

Going and class were not really tested. Their absence from the results is a fact about the scope of this series (four All-Weather courses, class 2 to 5 handicaps, 2006 to 2015), not evidence that either is uninformative.

Field size is the only race characteristic with real coverage. The winning band, 11 to 12 runners, is 23.5% of the search set, and a rule on it bets 357 times on test. That is not enough for its ROI to mean much either way.

The place prices are a construction and the each-way market is not a market. Set out in Section 5.

The forward strategy is untested. Section 5 reads paper 4’s result alongside a market-structure argument. Its two load-bearing assumptions, that usable size exists at early prices and that the model beats an early traded price, are not tested anywhere in this series and cannot be tested on this archive.

Harville, David A. 1973. “Assigning Probabilities to the Outcomes of Multi-Entry Competitions.” *Journal of the American Statistical Association* 68 (342): 312–16.

Lo, Victor S. Y., and John Bacon-Shone. 1994. “A Comparison Between Two Models for Predicting Ordering Probabilities in Multiple-Entry Competitions.” *Journal of the Royal Statistical Society: Series D (The Statistician)* 43 (2): 317–27.

Owen, Alun. 2019. “Statistical Models of Horse Racing Outcomes Using R.” *MathSport International 2019 Conference Proceedings* (Athens, Greece).